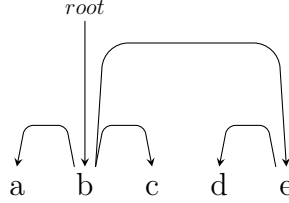


COMPUTER SCIENCE TRIPOS Part IB 75%, Part II 50% – 2020 – Paper 7

4 Formal Models of Language (pjb48)

The string ‘a b c d e’ has the following dependency parse:



The table shows the first two actions when deriving this parse using a modified shift-reduce parser which uses the actions: SHIFT, RIGHT-ARC and LEFT-ARC.

STACK	BUFFER	ACTION
	abcde	SHIFT
a	bcde	SHIFT
ab	cde	

dependency
parser actions

- (a) Describe the actions SHIFT, RIGHT-ARC and LEFT-ARC in terms of their effect on the stack and buffer. Also describe any arcs that are recorded when the actions are used. [5 marks]

Answer: For illustration, consider $\mathcal{L}(G_{dep}) \subseteq \Sigma^*$; during parsing the stack may hold γab where $\gamma \in \Sigma^*$ and $a, b \in \Sigma$, and b is at the top of the stack:

- (i) The SHIFT moves items of the input string from the buffer to the stack (1 mark)
- (ii) LEFT-ARC reduces the stack to γb and records use of rule $b \rightarrow a$ (2 marks)
- (iii) RIGHT-ARC reduces the stack to γa and records the use of rule $a \rightarrow b$ (2 marks)

producing a
dependency parse

- (b) The parser also uses the action TERMINATE which links a lone item on the stack to *root*. Complete the parse action table such that the actions derive the parse shown at the top of the page. [6 marks]

	STACK	BUFFER	ACTION	RECORD
		abcde	SHIFT	
	a	bcde	SHIFT	
	ab	cde	LEFT-ARC	$b \rightarrow a$
	b	cde	SHIFT	
<i>Answer:</i>	bc	de	RIGHT-ARC	$b \rightarrow c$
	b	de	SHIFT	
	bd	e	SHIFT	
	bde		LEFT-ARC	$e \rightarrow d$
	be		RIGHT-ARC	$b \rightarrow e$
	b		TERMINATE	$root \rightarrow b$

Using left-arc correctly (1 mark), using right-arc correctly (1 mark), using shift correctly (1 mark), shifting twice when required (1 mark), using terminate correctly (1 mark), completing

— *Solution notes* —

whole parse correctly (1 mark). The “record” column is provided for illustration and is not required.

parse action
selection

- (c) In Part (b) you have selected the parse actions by hand; explain how the sequence of parse actions is obtained from data in an automatic parser. [4 marks]

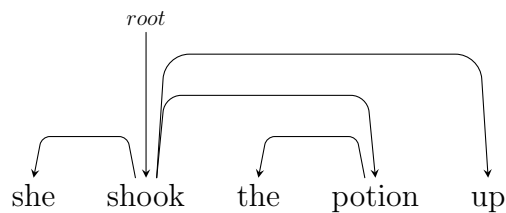
Answer: Mention of a dependency bank (1 mark), mention of a classifier (1 mark), mention of features (1 mark), some description of those features (1 mark).

word-order in
natural language

- (d) The dependency parse for the string ‘a b c d e’ is homomorphic with the English language string ‘Alice shook up the potion’. An alternative word order for this string is ‘Alice shook the potion up’.

- (i) Draw the dependency parse for the alternative string. [1 mark]

Answer:



-
- (ii) Compare capturing natural language alternations using a context-free grammar as opposed to a dependency grammar. [4 marks]

Answer: Relevant points not limited to: some description of a CFG (1 mark), constituency expressed through non-terminals (1 mark), two rules required to express this alternation (1 mark), many rules required for languages with flexible word ordering (1 mark).
